KALPANA J

FULL STACK DEVELOPER | AI-POWERED WEB APPLICATIONS

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Portfolio https://github.com/KalpanaJayapaul/KalpanaJayapaul-Portfolio

PROFESSIONAL SUMMARY

Full Stack Developer with 1.5+ years of hands-on experience in designing, developing, and maintaining responsive web applications, RESTful APIs, and administrative dashboards. Proficient in React.js, Angular, JavaScript, TypeScript, HTML5, CSS3, Python, Django REST Framework, PHP, and Laravel, with experience in SQL and MongoDB databases. Skilled in developing frontend and backend solutions, implementing CRUD operations, authentication, and API integrations. Experienced in integrating AI-powered applications using OpenAI, Gemini, and Groq APIs, including LLM integration, prompt engineering, and voice-based workflows. Passionate about building scalable, user-friendly applications and delivering efficient software solutions.

TECHNICAL SKILLS

Frontend: React.js, Angular, TypeScript, HTML5, CSS3, Material UI (MUI), Clarity UI
Backend: Python, Node.js, Django REST Framework, Core PHP, Laravel, REST APIs
Databases: MySQL, PostgreSQL, MongoDB, SQL
AI & Automation: OpenAI API, Gemini API, Groq API, LLM Integration, Prompt Engineering, AI Agents, Deepgram, Vosk, Speech-to-Text (STT), Text-to-Speech (TTS)
Cloud & Developer Tools: AWS SES, Git, GitHub, Postman, PyCharm
Other Technologies: WebSockets, BeautifulSoup4, CloudScraper

PROFESSIONAL EXPERIENCE

Full Stack Developer | Malaris Software Solutions | *Mar 2025 – Present*

- Develop and maintain responsive web applications and interactive user interfaces using React.js, Angular, TypeScript, JavaScript, and Material UI.
- Design, develop, and integrate RESTful APIs using Django REST Framework, PHP, and Laravel, implementing CRUD operations, authentication, and frontend-backend communication.
- Develop an AI-powered online clinic appointment booking application with chat and call assistance, integrating LLM APIs and knowledge-base management.
- Implement AI-driven voice workflows using Deepgram and Vosk, incorporating Speech-to-Text (STT), Text-to-Speech (TTS), and real-time communication capabilities.
- Integrate OpenAI and Groq APIs into applications, implementing prompt engineering and function calling to support intelligent AI-powered features.
- Develop and maintain a college administration and management system with modules for student and staff management, admissions, timetables, examinations, fees, courses, semesters, and batches.
- Build web applications using Core PHP, implementing frontend interfaces, backend business logic, database operations, and application functionality.
- Develop an online ticket-booking administration panel to manage ticketing operations and administrative workflows.
- Build a Laravel-based inventory and billing dashboard supporting stock management, inventory calculations, estimates, invoice generation, and business operations.
- Integrate AWS-based email services for sending and receiving application communications.
- Support AI/ML project development for internship students, including project setup, dataset preparation, model experimentation, and implementation guidance.

SELECTED PROJECTS & IMPLEMENTATIONS

AI Agent – Online Clinic Appointment Booking
Technologies: React.js, Angular, LLM APIs, OpenAI API

- Developed an AI-powered appointment booking workflow with chat/call assistance.
- Integrated LLM APIs and knowledge-base management.
- Implemented voice workflows using STT and TTS.

College Management / Admin Panel

Technologies: Angular, Laravel, PHP, SQL

- Developed administrative modules for student management, attendance, fees, and examination.
- Implemented CRUD operations and database integration.

Inventory, Estimates & Invoice Dashboard

Technologies: Laravel, PHP, SQL

- Developed inventory tracking and invoice generation functionality.

Online Ticket Booking – Admin Panel: Administrative interface for managing online ticket-booking operations and related workflows.

Core PHP Websites (5): Built five websites using Core PHP, including database-backed functionality and application-side logic.

AI/ML Projects for Internship Students: Implemented and supported AI/ML project work for internship students, including dataset preparation/model experimentation and project implementation support.

AWS Email Integration: Worked on AWS-based email sending and receiving functionality for application communication.

EDUCATION

B.E. Computer Science and Engineering — PSN College of Engineering and Technology | 2021–2025 | CGPA: 8/10

HSC — St. Anne's Girls Higher Secondary School, Chennai | 2020–2021 | 81.36%

SSLC — St. Anne's Girls Higher Secondary School, Chennai | 2018–2019 | 75%

INTERNSHIP EXPERIENCE

Bluon Technology, Kovilpatti | React.js Intern — May 2024

- Learned React.js fundamentals and Developed simple UI components.

Code Binds Technology, Coimbatore | Frontend Development Intern — September 2023

- Practiced HTML, CSS, and JavaScript.
- Developed frontend interface components.